THE UNITED REPUBLIC OF TANZANIA
THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT
ANNUAL JOINT EXAMINATION

FORM THREE
036 INFORMATION AND COMPUTER STUDIES

Time: 3 Hours **November 2024**

Instructions

1. This paper consists of sections A, B and C with a total of **twelve (12)** questions.
2. Answer **all** questions in sections A, B and **two (2)** questions in section C.
3. Section A carries **fifteen (16)** marks, section B **seventy (54)** marks and section C carries **fifteen (30)** marks.
4. Cellular phones and any unauthorized materials are **not** allowed in the examination room.
5. Write your **Examination Number** and **Full Name** on every page of your answer sheet(s).
6. All answers should be written with black or blue ink in the space provided.

SECTION A (16 Marks)

Answer **all** in questions in this section.

1. For each of the items (i) - (x), choose the correct answer from the given alternatives and write its letter beside the item number in the answer sheet provided.

(i) Which of the following best describes the concept of data entropy in information theory?
A The measure of the randomness or unpredictability of data
B The process of compressing data to reduces its size
C The mathematical formula for calculating the information content of data
D The principle that data should be organized in a hierarchical structure for efficient retrieval
E The process of providing a way to quantify the amount of surprise for an event measured in bits

(ii) A person who uses the computer skills to gain access of information illegally is known as
A Theft B Programmer C Hacker D administrator E Analyst

(iii)Why modern ways of disseminating information are preferred than traditional ways?
A They are expensive B They need skill C They are Complex
D They are fast E They are not selective

(iv) Which of the following element is responsible for making the text bold in HTML?
A <pre> B <a> C <b> D
 E <p>

(v) What is the blank space outside the printing area on a page?

A Clipart B Margins C Header

D Footer E Ruler

(vi) A term in computer terminology is a change in technology a computer is/was being used.

A Development B Version C Advancement

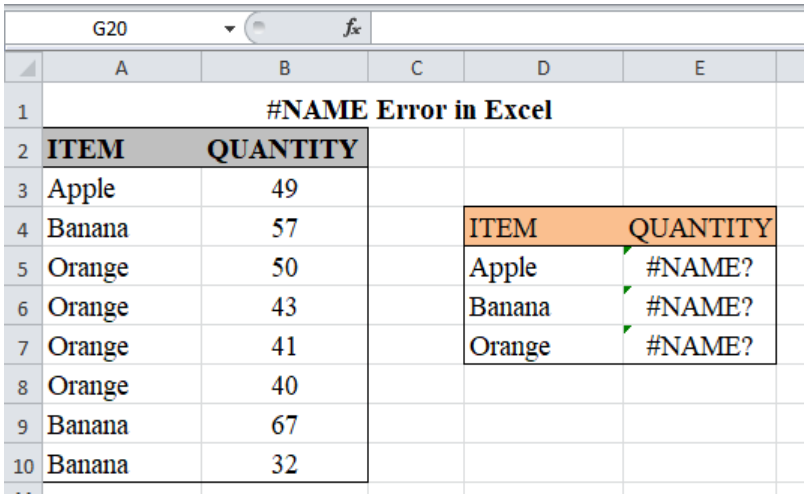
D Growth E Generation

(vii) Which of the following is not a cybercrime?

A Denial of Service B Man in the Middle C Malware

D Advanced Encryption Standard E Eaves dropping

(viii) The following spreadsheet was prepared to show the results of the students who did one different tasks. In cells E5 an error occurred. What might be the reason?



	A	B	C	D	E
1	#NAME Error in Excel				
2	ITEM	QUANTITY			
3	Apple	49			
4	Banana	57		ITEM	QUANTITY
5	Orange	50		Apple	#NAME?
6	Orange	43		Banana	#NAME?
7	Orange	41		Orange	#NAME?
8	Orange	40			
9	Banana	67			
10	Banana	32			

A There is invalid cell referencing in the formula used

B The column is not wide enough to accommodate the value

C The formula uses incorrect argument or operator

D The reference to a value is not available to a formula

E The formula references an invalid intersection of cells

(ix) User communicates with a computer with the help of which devices?

A Input device B Output device C Software device

D Both A and B E Central processing unit

(x) Form Two students were editing and modifying documents and appears very attractive characters through word processors in Computer laboratory. Which describe the above Form Two students work?

A Viewing B coloring C Formatting

D Layout E Styling

2. Match the following functions of each component in **List A** to their corresponding network device names in **List B** by writing the letter of the correct option beside the item number in the answer sheet provided.

List A	**List B**
i. A hardware device that connects at least two networks and forwards communication traffic between them. ii. A hardware device used to connect one network device to other network devices. iii. A hardware device that filters and forwards network traffic. iv. A hardware device that enables a computer to connect to a network. v. A hardware device used to connect multiple network devices together. vi. A hardware device that connects a computer to a broadband network.	A. Network Switch B. Network Modem C. Network Interface Card D. Network Hub E. Network Router F. Network repeater G. Network cable

SECTION B (54 Marks)

Answer **all** in questions in this section.

3.(a) What is an internet Services Provider

(b) A new company SEYPALM NEWS intends to go into the business of word processing. Advice the company on three computer-hardware system specifications features to consider as a measure of enhancing performance.

(c) You are provided with serial cable, USB cable, and parallel cable. Which of the cables provided can be used to.

i. Transmit data in a short distances
ii. Transit data in a long distance
iii. Support variety of peripheral devices

4. (a) Give the importance of backing up data.

(b) Briefly describe, the following terms as used with disk management

i. Formatting
ii. Compression
iii. Partitioning
iv. Fragmentation

5. (a) Explain the following e-mail address parts nyalulubogirls@gmail.com.

(b) i) Define the term Search engine

ii) Give two examples of Search engines

(c) Differentiate between the following terms.

i) Downloading and Uploading

ii) Internet and network
iii) User Log in and sign up

6. (a) One of the functions of an Operating System is job scheduling. Explain what is meant by job scheduling.
   (b) List and explain three types of user interfaces.
   (c) Describe the following categories of software
   i. Firmware
   ii. Proprietary software

7. (a) Define the term computer virus
   (b) Give any four sources of a computer virus
   (c) Explain any four control measures against viruses

8. Study the spreadsheet below and answer the questions that follow.

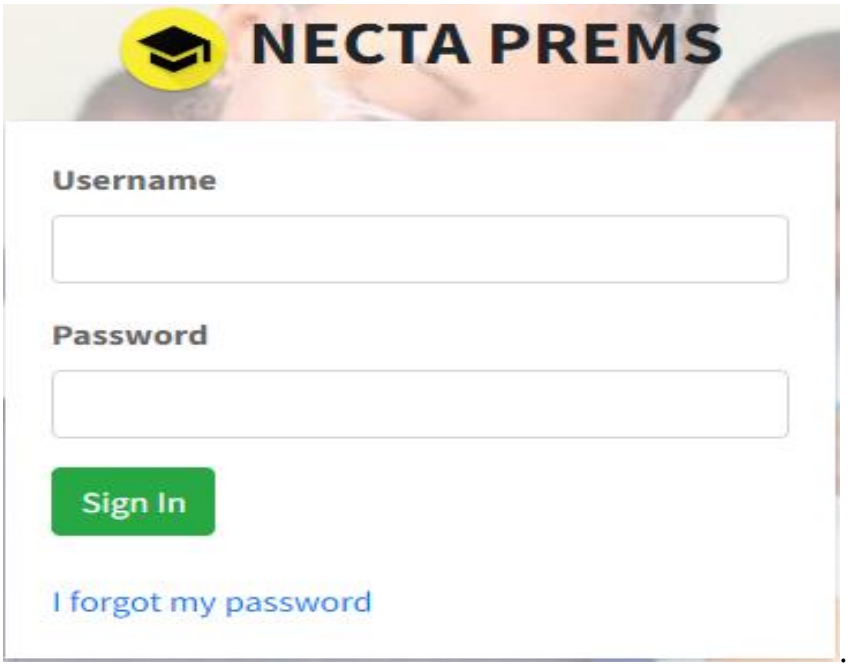
E11 f_x

	A	B	C	D
1	**WESTERN COMP BOOK CENTRE SALES**			
2	**BOOK TITLE**	**PRICE PER BOOK**	**BOOKS**	
3	D.BASE IV	400	145	
4	LOTUS FOR DUMMIES	460	15	
5	OFFICE WORD IN 3DAYS	300	65	
6	LEARN HTML 3DAYS	700	100	
7	TEACH YOURSELF JAVA	700	200	
8	COMPUTER STUDIES	500	300	
9	THE CLEVER FOOL COMPUTER	300	10	

   i. Write down the formula that can be used to find the price of the most costly book.
   ii. Write down the formula that can be used to determine the total sales for the book titled COMPUTER STUDIES
   iii. Write down the formula that can be used to determine the average of price of the books.
   iv. Write down formula in cell D6 that can be used to find the new price per book. If went up by percentage written if cell C4 and the formula has to be entered only in cell D3 then copied to other cells.
   v. Write down the output total sales in D7 if B6 is 10%.

9. (a)Write the long form to each of the following acronyms as used in ICS (04 marks)
   i. BIOS
   ii. URL
   iii. WWW
   iv. HDMI

(b) Write the HTML codes that will display the following webpage interface on a browser



HINT:

- Use the NECTA picture which named **logo1.jpeg**.
- Use "**I forgot my password**" to add link with file named **forgotfile** access user account.

SECTION C (30 Marks)

Answer **two (2)** questions in this section.

10. Despite the ICT advancements and contributions to socio-economic development, there are threats brought about by its use. Such threats trigger the need for security mechanisms. What are the four common security measures practiced to protect information.

11. When computers and other electronic devices are integrated in teaching and learning process, studying becomes more engaging for students. How do Information and Computer Studies relate to other subjects such Physics, Chemistry and Biology? Give four points.

12. Elaborate any four advantages of using electronic spreadsheets as compared to traditional worksheets.